

Michelle Zuo

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EDUCATION

Duke University

Mathematics and Computer Science Major

Durham, NC

Aug. 2026 – May 2030

Thomas Jefferson High School for Science and Technology

Weighted GPA: 4.589, AP Scholar with Distinction, AIME Qualifier (4x)

Alexandria, VA

Aug. 2023 – June 2026

EXPERIENCE

Data Analyst

Duke Impact Investment Group (DIIG)

Sep. 2026 – Present

Durham, NC

- Prioritized 9 research questions into a top-5 roadmap for ADYN (YC25) by aligning business and scientific goals
- Analyzing 7 datasets (orders, hormone, genotype) with a 5-person team via documented EDA and cleaning
- Building a customizable dashboard demo so ADYN staff can self-serve insights, backed by a metrics data dictionary

Operations Team Member

HackDuke

Sep. 2026 – Present

Durham, NC

- Raising sponsorship funding for Duke's premier hackathon by contacting and pitching prospective sponsors
- Managing event logistics for a flagship campus hackathon by coordinating with the operations team
- Growing sponsor relationships by sustaining outreach and partner communications across the planning cycle

Research Intern, NSF-Funded DISES Project

George Mason University (ASSIP)

June 2025 – March 2026

Fairfax, VA

- Correlated economic loss with social vulnerability across NOAA's top 10 billion-dollar hazards to guide disaster aid
- Processed terabyte-scale datasets using Python and GeoDa via statistical modeling and spatial visualization
- Co-authoring a multi-hazard disaster loss paper, currently under review for journal submission

Research Student

Institute for Mathematics and Democracy (MAA-Sponsored)

June 2025 – March 2026

Remote

- Modeled Wisconsin election data with Möbius inversion and machine learning to analyze swing-state behavior
- Authored a paper preprinted on arXiv (2510.26867), under review at Rose-Hulman Undergraduate Math Journal
- Presented findings to all program teams after a competitive national selection into the 5-week program

Quantum Research Student

Virginia Tech Quantum Lab Partnership

Sep. 2025 – June 2026

Virginia

- Improving ADAPT-VQE, a quantum algorithm for molecular simulation, by leveraging entanglement in Python
- Iterated on algorithm design with a faculty mentor and 2 fellow researchers through bi-weekly research meetings

PROJECTS

FindAQuire | *Python, Flask, Grover's Algorithm, HTML*

Mar. 2026

- Built a full-stack app that ranks antibody-antigen docking sites from FASTA input to speed up antibody discovery
- Applied Grover's quantum search for a quadratic speedup, $O(N)$ to $O(\sqrt{N})$, over classical docking search
- Validated the pipeline end-to-end on an HIV-1 antigen/antibody pair from RCSB PDB via Flask

TECHNICAL SKILLS

Languages: Python, Java, R, MATLAB, LaTeX, Chinese

Frameworks & Tools: Flask, GeoDa

Focus Areas: Quantum Computing, Machine Learning, Statistical Modeling, Spatial Analysis, Data Visualization